Name:__________________ Class:_______ Student No.:______ Phone No.:______________

香港數學奧林匹克學校 主辦

Hong Kong Mathematical Olympiad School

2023 The 30th Hong Kong Primary Mathematical Olympiad

Remarks:

(1). 1 mark for each question, total 20 marks. Each mark is only given to exactly right answer.

(2). Write the answers in the space provided, otherwise the answer will not be marked.

(3). Calculate the following questions. Answers expressed in fraction should be expressed in simplest form, otherwise will not be marked.

Write down 0 to 9 in the following table, to help the marker know your handwriting.

0	1	2	3	4	5	6	7	8	9

Marks:

1. $1122+1212+1221+2112+2121+2211=?$ 1.________

2. $23.5\times12+0.15\times120+2500\times0.12=$ 2.________

3. $(\frac{1}{2}+\frac{2}{3}+\frac{3}{4})\times(1+\frac{1}{2}+\frac{2}{3}+\frac{3}{4})-(\frac{1}{2}+\frac{2}{3}+\frac{3}{4}-1)\times(2+\frac{1}{2}+\frac{2}{3}+\frac{3}{4})=?$ 3.________

4. $\frac{13}{26}+\frac{26}{52}+\frac{52}{104}+\ldots+\frac{1664}{3328}=?$ 4.________

5. Denote $S(n)$ to be the sum of all digits of n, e.g. $S(1234)=1+2+3+4=10$. Find $S(11\times11)+S(111\times111)+S(1111\times1111)$. 5.________

6. $\dfrac{2\times6\times10+4\times8\times18+6\times12\times20}{1\times3\times5+2\times4\times9+3\times6\times10}=?$ 6.________

7. By observing the following pattern, from the 1st row to the 50th row, how many multiples of 3 are there? 7.________

1st row：1

2nd row：1 2

3rd row：1 2 3

…

50th row：1 2 3 … 48 49 50

8. Define $A\#B=A^2+2\times A\times B+B^2$. Find $3\#4+4\#5+5\#6-5\#4-4\#3$. 8.________

9. Calculate the unit digit of $1\times5+5\times9+9\times13+\ldots+93\times97$. 9.________

10. In the right column form, different letters represent different numbers. Find the 7-digit number ABCDEFG.

$$
\begin{array}{ccccc}
 & A & B & C & D \\
\times & & & E & F \\
\hline
7 & B & 0 & B & \\
G & A & E & B & 0 \\
\hline
D & C & E & D & 0
\end{array}
$$

10.

11. Yan's present age is half of her mother. She remembers that when she first learnt Olympiad mathematics 15 years ago, her mother's age was 5 times of hers. How old is Yan now? 11.________

12. All the faces of a pentagonal prism are coloured such that two adjacent faces sharing a common edge are with different colours. Find the minimum number of colour is needed. 12.________

13. In a take away platform, Wang's restaurant offers discount to customers. A 20% discount will be offered to those orders with $150 or above. It is known that the prices of pork chop rice, beef rice, chicken leg rice are $69, $71 and $84 respectively. Milk tea is $10. Jet likes to order two different kinds of rice. Milk tea is optional. How much should he pay at least? 13. $________

14. Town A and B are connected by a straight road. A truck and a motorcycle start from A and B respectively, are travelling towards each other with speed ratio $3:5$. They meet 20min after starting. How long does the truck take to go to Town B after they meet? 14.________ min

15. It costs Ming $87 to buy 2 oranges, 5 pears and 8 peaches. It also costs him 1 orange, 2 pears and 3 peaches for $34. How much does he pay for 1 orange, 4 pears and 7 peaches? 15. $________

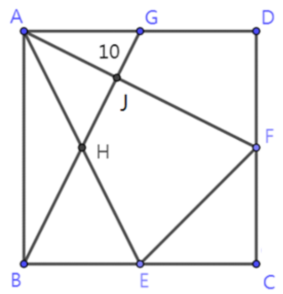
16. Consider the sum of 2 numbers which are selected among 7 numbers 4, 7, 10, 11, 22, 34 and 37. Find the number of combinations such that this sum is a multiple of 4. 16.________

17. In a fruit market, each box contains 795 oranges. There are 92342 boxes. If every 9 oranges are packed together, how many oranges are remained? 17.________

18. There is less than 45 students in a class. Some new students join the class, and the number of students increase 12.5%. Find the maximum possible number of students in the class originally. 18.________

19. In a pizza shop, customer Ming orders a pizza with radius 10 inches. However, the shop just has 3 inches and 4 inches radius pizzas which are with the same thickness as the ordered pizza. If the same amount of pizza is provided, how many pizzas should be offered? (Take $\pi = 3.14$ if necessary.) 19.________

20. As shown in the figure, *ABCD* is a square. *E*, *F* and *G* are the mid-point of *BC*, *CD* and *AD* respectively. The area of ΔAGJ is 10cm^2. Find the difference in area between quadrilaterals *EFJH* and *FDGJ*. 20.________ cm^2



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